Western Science **Department of Computer Science**

CS3307A Form B: Contribution Statement

Submitted with Deliverable 2 and with Deliverable 3, by every team

Object-Oriented Design & Analysis · Fall 2026 · Instructor: Dr. Umair Rehman

Who	Every team: one form per team for each of Deliverables 2 and 3. A pair completes Sections 1 to 4; a student working individually completes Sections 1, 3, and 4.
When	With Deliverable 2 and with Deliverable 3, as a separate PDF on Brightspace and in the `report/` folder of the repository.
How	Fill in these pages or reproduce them in your own document, keeping the headings, questions, and declaration. A typed full name is accepted as a signature.

Project title and group number	Mirage: Casino Game , Group Number: 162
Deliverable	☑ Deliverable 2 ☐ Deliverable 3
Arrangement	☐ Individual ☑ Pair
Member(s)	Liam Wray, Abdul Hasna

1. What was done, and by whom (everyone)

List the main pieces of work in this deliverable. Be specific: name classes, modules, diagrams, report sections, and tests. In a pair, mark who did each item, marking both for work done together. An individual lists the work and leaves the member columns blank.

Work item (class, module, diagram, report section, tests, video)	Member 1	Member 2
blackjackwindow, pokerwindow, mainwindow, GameWindow	Liam	
Blackjack, Poker, IGame, GameManager, GameFactory	Liam	
blackjackwindow.ui, pokerwindow.ui, mainwindow.ui	Liam	
baccaratwindow, Baccarat, baccaratwindow.ui		Abdul
Animation classes	Liam	
UML Diagram, Use Case Diagram, Deliverable 2 report	Liam	
Sequence Diagram, Deliverable 2 Report		Abdul

2. Overall split (pairs only)

Our agreed estimate of each member's share of the total effort on this deliverable (the two figures sum to 100%):

Member 1: 70 % Member 2: 30 %

If the split is more uneven than 60/40, or if it differs noticeably from what the commit history shows (for example because you programmed together at one computer), explain briefly:

Because of our midterm schedules, we knew going into Deliverable 2 that Member 2 (Abdul) would be much busier than Member 1 (Liam). Therefore we agreed that Liam would have more responsibility for Deliverable 2 and Abdul would be in charge for Deliverable 3.

3. Outside help and third-party material (everyone)

List any help received from people outside the team (other than the instructor and TAs) and any third-party code, libraries, or assets included in the submission, with where they are used. Use of generative AI is reported on Form C, not here. Write "none" if there was none.

Dr. Umair's lectures were used to reinforce reasons to use certain patterns such as the Factory Pattern and Flyweight Pattern. The C++ code from those lectures was also used as a reference on how to implement the Factory and Flyweight Patterns.

For specific syntax issues we found while implementing GameFactory I consulted resources such as StackOverflow and GeeksforGeeks when asking about how to implement the Qt windows and for syntax specifications when making buttons.

4. How the work went (everyone)

What worked well, what did not, and any change to the plan recorded on Form A. In a pair, comment on the collaboration; an individual, on keeping to the internal deadlines.

The work went smoothly, we both had a clear understanding of our separate responsibilities and completed the work before our internal deadline of Monday November 9th.

Declaration

I/we confirm that this statement is accurate; that, apart from the material acknowledged in Section 3 and on Form C, the submission is my/our own work; that each member has reviewed the whole submission; and that each member can explain the design and the code submitted under their name. In a pair, a member who does not agree with this statement should not sign and should instead email the instructor before the 48-hour window closes.

Member 1 signature: Liam Wray Member 2 signature (pairs): Abdul Hasna

Date: November 8th 2026 Date: November 13th 2026